

Electric flux: Electric flux is the rate of flow of the electric field through a given area. If the electric field is uniform, the electric flux passing through a surface of vector area A is

$$\varphi_E = \vec{E} \cdot \vec{A} = EA \cos \theta$$

Consider infinitesimal surface element da of surface A as shown in fig. 1. Let us calculate the electric flux $d\varphi_E$ due to this element. We can write

$$d\varphi_E = E \cos \alpha da$$

Where α is the angle between $\vec{E}$ and unit vector $\hat{n}$ drawn along a normal to the surface. In vector

$$d\varphi_E = \vec{E} \cdot \hat{n} da = \vec{E} \cdot d\vec{a}$$

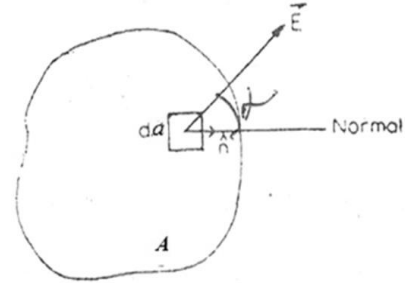


Fig. 1.

The flux due to entire closed surface is obtained by integrating the above equation around the surface, i.e.

$$\varphi_E = \oint_S d\varphi_E = \oint_S \vec{E} \cdot d\vec{a}$$

Unit: The unit of flux is Newton-meter²/Coulomb (N-m²/C).

Gaussian surface: A hypothetical closed surface of any shape drawn in an electric field for the purpose of solving problems concerning electric flux is called Gaussian surface. The shape of the Gaussian surface is decided on the basis of symmetry of the problem so that the value of electric flux $\varphi_E = \oint_S \vec{E} \cdot d\vec{a}$ can be calculated.

Gauss's law

Statement: The flux of the electric field $\vec{E}$ through any closed surface, i.e. the integral $\oint \vec{E} \cdot d\vec{a}$ over the surface, is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed by the closed surface. If the surface does not enclose the charge, the flux of $\vec{E}$, i.e. $\oint \vec{E} \cdot d\vec{a}$ equals zero.

Thus we can write

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}, \quad \text{when } q \text{ is inside the surface} \quad \text{or} \quad \epsilon_0 \oint_S \vec{E} \cdot d\vec{a} = q$$

$$\oint_S \vec{E} \cdot d\vec{a} = 0, \quad \text{when } q \text{ is outside the surface} \quad \text{or} \quad \oint_S \vec{E} \cdot d\vec{a} = 0$$

Proof: Suppose a positive charge q is enclosed in a surface of arbitrary shape. The electric field $\vec{E}$ at every point of the surface is directed radially outward from the charge, and its magnitude at the point r (Fig. 2.) is

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

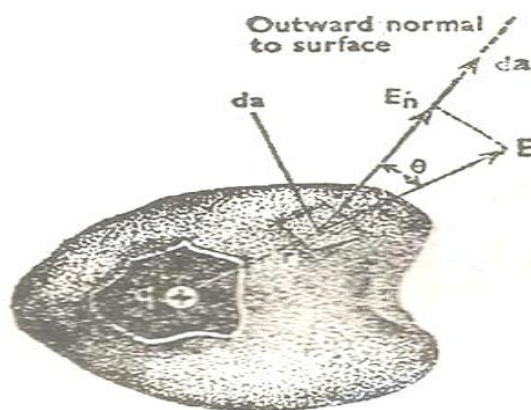


Fig. 2. An imaginary closed surface which encloses a point charge.

Over an infinitesimal area da the magnitude and the direction of the field remains the same.

The component $\vec{E}$ along the normal is $E_n = E \cos \theta$, where θ is the angle between $\vec{E}$ and the outward normal to the surface.

Hence

$$E_n da = E \cos \theta da = \frac{q \cos \theta da}{4\pi\epsilon_0 r^2} \quad (1)$$

$\frac{da \cos \theta}{r^2} = d\Omega$ is the element of solid angle subtended by da at the point where q is located.
Thus

$$E_n da = E \cos \theta da = \frac{q}{4\pi\epsilon_0} d\Omega \quad (2)$$

Regardless the shape and the size of the closed surface.

$$\int d\Omega = 4\pi \text{ steradians}$$

Hence integrating over the entire closed surface equation (2) yields

$$\oint_S E_n da = \frac{q}{4\pi\epsilon_0} \oint d\Omega = \frac{q}{\epsilon_0} \quad (3)$$

Where $\oint_S$ means closed surface integral. Equation (3) can be put in a vector notation if we note that the vector element area $d\vec{a}$ is along the normal. Hence, $E_n da = \vec{E} \cdot d\vec{a}$. Thus equation (3) takes the form

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0} \quad (4)$$

If the charge q is negative then the direction of $\vec{E}$ would be radially inward. But, since the direction of $d\vec{a}$ (assumed to be positive in the outward direction) remains unchanged, equation (4) is reproduced again.

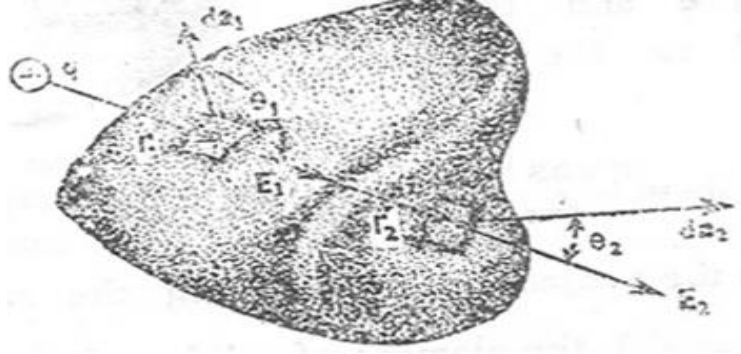


Fig. 3. A point charge outside of an imaginary closed surface.

Now consider the charge q lying outside the closed surface as shown in fig. 3. The direction of the field due to the charge q is inward at $\vec{r}_1$ on area da_1 and outward at $\vec{r}_2$ on area da_2 . But the solid angle subtended by da_1 and da_2 at q are the same. In other words

$$d\Omega = \frac{da_1 \cos \theta_1}{r_1^2} = \frac{da_2 \cos \theta_2}{r_2^2} \quad (5)$$

$$\text{But} \quad \vec{E}_1 \cdot d\vec{a}_1 = -E_1 da_1 \cos \theta_1 = -\frac{q}{4\pi\epsilon_0} \frac{da_1 \cos \theta_1}{r_1^2}$$

$$\text{and} \quad \vec{E}_2 \cdot d\vec{a}_2 = E_2 da_2 \cos \theta_2 = \frac{q}{4\pi\epsilon_0} \frac{da_2 \cos \theta_2}{r_2^2}$$

$$\text{Thus} \quad \vec{E}_1 \cdot d\vec{a}_1 + \vec{E}_2 \cdot d\vec{a}_2 = \frac{q}{4\pi\epsilon_0} \left[\frac{da_2 \cos \theta_2}{r_2^2} - \frac{da_1 \cos \theta_1}{r_1^2} \right] = 0 \quad (6)$$

Further, it is evident from fig. 3. that whole closed surface can be divided into pairs of such elementary areas and the positive and negative contribution on the pair due to a charge lying outside closed surface will mutually cancel and, therefore the sum total contribution due to it on the closed surface will be zero. Thus

$$\oint_S \vec{E} \cdot d\vec{a} = 0 \quad (7)$$

if the charge q lies outside the closed surface.

Finally, we may have more than one charges inside the closed surface. In that case we are to take the surface integrals for individual charges and add them up to get

$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{a} = \sum_{i=1}^n q_i \quad (8)$$

Where the summation on the right side of equation (8) is to be carried out over all charges inside the closed surface. If, on the other hand, the charges lie outside the closed surface we would get equation (7), since the contribution of each charge would then vanish.

Applications of Gauss's law

(i) Deduction of Coulomb's law

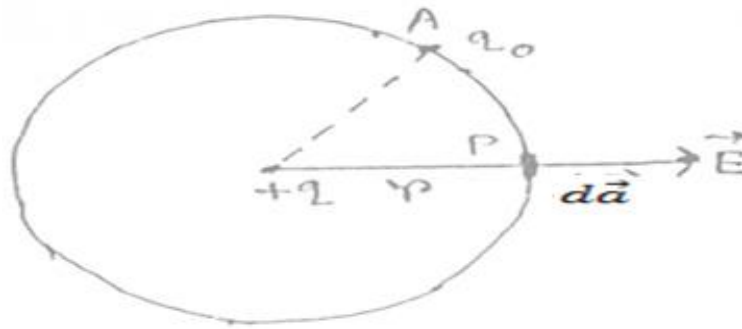


Fig. 4

Coulomb's law can be deduced from Gauss's law. Let us consider a Gaussian surface with radius r and the charge q is at the centre (Fig. 4). For symmetry at any point on the Gaussian surface both $\vec{E}$ and $d\vec{a}$ are directed radially outward. So that, at any point P on the Gaussian surface the angle between them is zero and $\vec{E} \cdot d\vec{a}$ becomes simply $E da$. Thus Gauss's law reduces to

$$\oint_S \vec{E} \cdot d\vec{a} = \oint_S E da = \frac{q}{\epsilon_0} \quad (1)$$

Since E is constant on the surface, it can be taken outside the integral sign, and consequently from equation (1)

$$\oint_S E da = E \oint_S da = \frac{q}{\epsilon_0}$$

$$\text{Or} \quad E(4\pi r^2) = \frac{q}{\epsilon_0}$$

$$\therefore E = \frac{q}{4\pi\epsilon_0 r^2}$$

Let us put a second point charge q_0 at the point A where $\vec{E}$ is calculated. The magnitude of the force that acts on

$$F = q_0 E$$

$$F = q_0 \frac{q}{4\pi\epsilon_0 r^2}$$

$$\therefore F = \frac{qq_0}{4\pi\epsilon_0 r^2} \quad (2)$$

Equation (2) is the Coulomb's law in electrostatic force. Thus Coulomb's law can be deduced from Gauss's law.